

State Estimation with Kalman Filter and Moving Horizon Estimation on the Temperature Control Lab

Johannes Stockhammer

*Otto-von-Guericke University Magdeburg, Laboratory for Systems Theory and Automatic Control,
Magdeburg, Germany*

Abstract

This report applies and compares different state estimators on the Temperature Control Lab (TCLab). The standard, extended and unscented Kalman filter (SKF, EKF, UKF) and Moving Horizon Estimation (MHE) are implemented in simulation and in real time, and are used to estimate temperatures and a model parameter of the TCLab. In single-heater operation, SKF, EKF and UKF are compared. In dual-heater operation, EKF, UKF and MHE are studied. On the real system, the model mismatch leads to a noticeable estimation error of the second heater temperature, and the model does not describe the dynamic behaviour correctly. To show a way to reduce this mismatch, MHE is used to estimate the heat capacity of the heaters. A further investigation and correction of the model is needed to improve its dynamic behaviour.

Keywords: Temperature Control Lab, Kalman filter, Unscented Kalman filter, Moving Horizon Estimation

1. Introduction

The history of the Kalman filter (KF) starts in 1960, when Rudolf Emil Kalman published his paper “A new approach to linear filtering and prediction problems” [1]. In it, he describes an optimal estimator and filter for linear, discrete-time systems that estimates states on the basis of noisy measurements. The filter works recursively and is based on a stochastic description of the process and measurement noise.

The best-known application of the Kalman filter is the Apollo Guidance Computer (AGC), where it was a key element of the moon landing in 1969. The flight software of Apollo 11 (Luminary 099) is publicly available [2] and contains a routine named KALMAN_FILTER. Because of model errors and the strongly nonlinear dynamics of the spacecraft, the original approach was modified and implemented as the Kalman–Schmidt filter, today known as the Extended Kalman Filter (EKF) [3].

This report applies three variants of the Kalman filter and Moving Horizon Estimation to the Temperature Control Lab, a small thermal laboratory system, first in simulation and then on the real hardware.

Course project report, Otto-von-Guericke University Magdeburg, 2021. Revised edition 2026.

DOI: 10.5281/zenodo.22984395

© Johannes Stockhammer. Licensed under CC BY 4.0, except third-party contents (see the end of the report).

2. Theory: Kalman Filter

The Kalman filter is an optimal estimator for linear processes in state-space form with Gaussian noise. Modified versions also handle moderately nonlinear systems. Provided that the process noise covariance Q and the measurement noise covariance R are known, the filter uses the noisy process, the noisy measurements and the noise-free control input u to obtain a better estimate of the state x . In contrast to a conventional low-pass filter, the estimate of the Kalman filter has no phase shift. This report uses three variants of the filter, all in discrete time.

The standard Kalman filter (SKF) is limited to linear systems. It consists of two equations for the prediction and three equations for the correction step. First, the state $\hat{x}_k^{(-)}$ is predicted from the previous estimate $\hat{x}_{k-1}^{(+)}$ and the control input u_{k-1} . Then the covariance of the prediction error $P_k^{(-)} = \text{Var}(x_k - \hat{x}_k^{(-)})$ is calculated from the previous covariance $P_{k-1}^{(+)}$, the process noise matrix G_{k-1} and the process noise covariance Q [4].

Prediction

$$\hat{x}_k^{(-)} = A_{k-1} \hat{x}_{k-1}^{(+)} + B_{k-1} u_{k-1} \quad (1a)$$

$$P_k^{(-)} = A_{k-1} P_{k-1}^{(+)} A_{k-1}^T + G_{k-1} Q_{k-1} G_{k-1}^T \quad (1b)$$

In the correction step, the Kalman gain K_k is calculated from the prediction error covariance $P_k^{(-)}$, the output matrix C_k and the measurement noise covariance R_k . The gain weights the difference between the measurement y_k and the predicted output $C_k \hat{x}_k^{(-)}$; the weighted difference is added to the predicted state. The corrected covariance $P_k^{(+)}$ follows from the predicted covariance, the Kalman gain and the output matrix.

Correction

$$K_k = P_k^{(-)} C_k^T (C_k P_k^{(-)} C_k^T + R_k)^{-1} \quad (2a)$$

$$\hat{x}_k^{(+)} = \hat{x}_k^{(-)} + K_k (y_k - C_k \hat{x}_k^{(-)}) \quad (2b)$$

$$P_k^{(+)} = P_k^{(-)} - K_k C_k P_k^{(-)} \quad (2c)$$

The extended Kalman filter (EKF) handles nonlinear models by linearising them around the current estimate [5].

Prediction

$$\hat{x}_k^{(-)} = f(\hat{x}_{k-1}^{(+)}) \quad (3a)$$

$$\hat{y}_k = h(\hat{x}_k^{(-)}) \quad (3b)$$

$$P_k^{(-)} = \frac{\partial f}{\partial x} \bigg|_{\hat{x}_{k-1}^{(+)}} P_{k-1}^{(+)} \frac{\partial f}{\partial x} \bigg|_{\hat{x}_{k-1}^{(+)}}^T + G_{k-1} Q_{k-1} G_{k-1}^T \quad (3c)$$

Correction

$$P_{xy} = P_k^{(-)} \left. \frac{\partial h}{\partial x} \right|_{\hat{x}_k^{(-)}}^T \quad (4a)$$

$$P_{yy} = \left. \frac{\partial h}{\partial x} \right|_{\hat{x}_k^{(-)}} P_k^{(-)} \left. \frac{\partial h}{\partial x} \right|_{\hat{x}_k^{(-)}}^T + R_k \quad (4b)$$

$$K_k = P_{xy} P_{yy}^{-1} \quad (4c)$$

$$\hat{x}_k^{(+)} = \hat{x}_k^{(-)} + K_k (y_k - \hat{y}_k) \quad (4d)$$

$$P_k^{(+)} = P_k^{(-)} - K_k P_{yy} K_k^T \quad (4e)$$

The unscented Kalman filter (UKF) handles nonlinear models with the unscented transform: a small set of sigma points is propagated through the nonlinear functions, and mean and covariance are calculated from the transformed points [6].

Observability and controllability. Of a general system, only the control inputs u and the outputs y are directly accessible; the states are internal. The Kalman filter requires the system to be observable: every state x can be determined from the input $u(t)$ and the output $y(t)$ within a finite time. Controllability is the dual property: a suitable input $u(t)$ can transfer the system from any initial state to any final state within a finite time. Consider the linear state-space model in continuous and discrete time:

$$\begin{aligned} \dot{x}(t) &= A x(t) + B u(t) & x_{k+1} &= A_d x_k + B_d u_k \\ y(t) &= C x(t) + D u(t) & y_k &= C x_k + D u_k \end{aligned}$$

According to Kalman's criterion, a linear time-invariant system of order n is observable if the observability matrix $\underline{S}_B$ has rank n . For discrete-time systems, $\underline{S}_B^*$ has to be used: a system that is observable in continuous time need not be observable at the sampling instants of its discrete-time description [7].

$$\underline{S}_B = \begin{bmatrix} \underline{C} \\ \underline{C} \underline{A} \\ \underline{C} \underline{A}^2 \\ \vdots \\ \underline{C} \underline{A}^{n-1} \end{bmatrix} \quad \underline{S}_B^* = \begin{bmatrix} \underline{C} \\ \underline{C} \underline{A}_d \\ \underline{C} \underline{A}_d^2 \\ \vdots \\ \underline{C} \underline{A}_d^{n-1} \end{bmatrix} \quad (5a)$$

Accordingly, the system is controllable if the controllability matrix $\underline{S}_S$ (or $\underline{S}_S^*$ in discrete time) has rank n [7].

$$\underline{S}_S = [\underline{B} \quad \underline{A} \underline{B} \quad \underline{A}^2 \underline{B} \quad \cdots \quad \underline{A}^{n-1} \underline{B}] \quad (6a)$$

$$\underline{S}_S^* = [\underline{B}_d \quad \underline{A}_d \underline{B}_d \quad \underline{A}_d^2 \underline{B}_d \quad \cdots \quad \underline{A}_d^{n-1} \underline{B}_d] \quad (6b)$$

3. Theory: Moving Horizon Estimation

System models are often built from a combination of experimental knowledge and physical principles. On a real system, the parameters always deviate slightly from the model because of simplifications, measurement errors or inaccuracies. Moving Horizon Estimation (MHE) can estimate such parameters together with the states. MHE works on a horizon of fixed length: the number of states, parameters and measurements in the problem stays the same at every time step. When a new measurement arrives, the oldest measurement leaves the horizon. The principle is shown in Fig. 1.

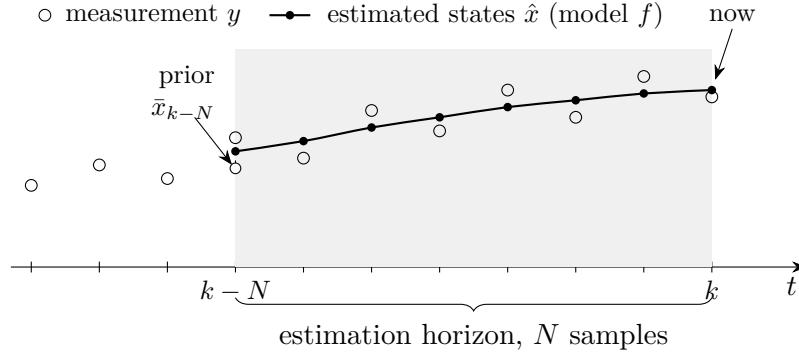


Fig. 1: Principle of Moving Horizon Estimation: within the horizon, the state trajectory is fitted to the measurements under the model constraints; the prior $\bar{x}_{k-N}$ carries the information of older measurements.

MHE is an optimisation-based estimator. It uses an internal model and the measurements within the horizon to estimate states, outputs or unknown parameters. Like every optimisation problem, MHE relies on a solver to find the minimum of the cost function. As in Model Predictive Control, a major benefit is the handling of soft and hard constraints, e.g. to keep temperatures, velocities or flow rates within physical limits. For nonlinear systems, the solver has to handle a nonlinear program (NLP). For strongly nonlinear systems, MHE can give better estimates than the EKF or the UKF.

MHE is based on *maximum a posteriori* estimation, i.e. on the mode of the probability density function. The EKF, in contrast, approximates the *a posteriori* distribution by a normal distribution, whose mean and mode coincide [8]. The drawback of MHE is its higher computational effort.

$$\{\hat{x}_{T-N+1}, \dots, \hat{x}_T\} = \arg \max_{x_{T-N+1}, \dots, x_T} p(x_{T-N+1}, \dots, x_T | y_0, \dots, y_T) \quad (7)$$

The objective function (cost function) defines the estimation problem. It states which deviations of measurements, states or parameters are penalised and how strongly, and which quantities are estimated. According to [9], a weighted squared error is the most common form. An example of an MHE cost function:

$$\begin{aligned}
& \underset{\hat{x}, \hat{p}}{\text{minimize}} && (y - \hat{y})^T P_y (y - \hat{y}) + (\bar{x} - \hat{x})^T P_x (\bar{x} - \hat{x}) + (p_n - p)^T P_p (p_n - p) \\
& \text{subject to} && 0 = f(\dot{x}, x, u, p), \\
& && 0 \leq g(\dot{x}, x, u, p)
\end{aligned} \tag{8}$$

with

$\hat{y}$	predicted model outputs	$f(\cdot)$	equality constraints (model)
y	measurements	$g(\cdot)$	inequality constraints
$\hat{x}$	estimated states	P_y	weight of the measurement residuals
$\bar{x}$	prior states (arrival cost)	P_x	weight of the state deviation
p	estimated parameters	P_p	weight of the parameter deviation
p_n	nominal parameters		

4. Methods

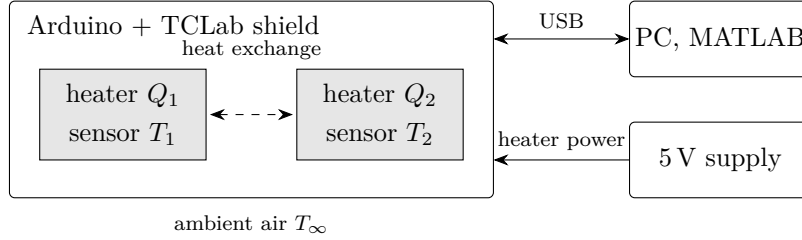


Fig. 2: Temperature Control Lab (TCLab): two transistor heaters with temperature sensors on an Arduino shield [10].

The practical part uses the Temperature Control Lab kit (TCLab) of APMonitor [10]. The work is done with MATLAB/Simulink R2021a and the MATLAB Support Package for Arduino Hardware; the software is installed according to the APMonitor guide [11]. With this package, the Arduino is connected with the MATLAB command $a = \text{arduino}(\text{com_port}, 'Uno')$. The MHE script and the recorded measurement data are available online [12].

The TCLab is an Arduino with a plug-on shield that carries two transistor heaters. Each heater has its own temperature sensor and heat sink, and the heater power of each heater is set individually. Because the heaters are close to each other, heat is exchanged between them and with the ambient air. The kit is designed for studying controller design and model identification. Software for Python and MATLAB/Simulink is available; this report uses MATLAB/Simulink only.

The aim of the work is to become familiar with state estimation and to learn how different variants of the Kalman filter and Moving Horizon Estimation perform in different applications. The standard Kalman filter (SKF), the extended Kalman filter (EKF) and the unscented Kalman filter (UKF) are used for state estimation; MHE is used for parameter estimation. The work is divided into three tasks.

4.1. Task 1

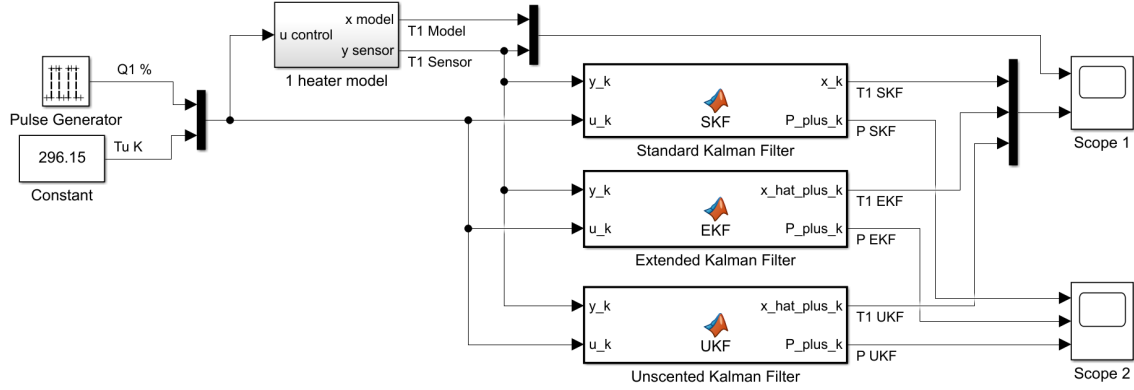


Fig. 3: Task 1, Simulink model *Model_1heater* (screenshot of MathWorks Simulink; not covered by the CC BY license of this report)

The single-heater model is taken from APMonitor [10]:

$$m c_p \cdot \frac{\partial T_1}{\partial t} = UA \cdot (T_\infty - T_1) + \varepsilon \sigma A \cdot (T_\infty^4 - T_1^4) + \alpha_1 Q_1 \quad (9)$$

with the parameters of Table 1. Rearranging Eq. (9) gives the nonlinear single-heater differential equation:

$$\dot{T}_1 = \frac{UA}{m c_p} \cdot T_\infty - \frac{UA}{m c_p} \cdot T_1 + \frac{\varepsilon \sigma A}{m c_p} \cdot T_\infty^4 - \frac{\varepsilon \sigma A}{m c_p} \cdot T_1^4 + \frac{\alpha_1}{m c_p} \cdot Q_1 \quad (10a)$$

$$= f_1(T_1, T_\infty, Q_1) \quad (10b)$$

The SKF is implemented for the general discrete-time case [4]. Therefore, the model (10) has to be linearised and discretised. The linearisation uses the first-order Taylor expansion (11) around the operating point (12).

$$T_{f(x;x_0)} = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} \cdot (x - x_0) \approx f(x_0) + f'(x_0) \cdot (x - x_0) = \dot{T}_{1_{approx}} \quad (11)$$

$$T_{\infty 0} = T_{10} = T_{20} = 23^\circ \text{C} = 296.15 \text{ K} \quad (12a)$$

$$Q_{10} = Q_{20} = 0 \text{ W} \quad (12b)$$

The operating point has to be an equilibrium: the constant term (13) of the Taylor expansion must be zero. Otherwise the linearised model cannot be written in state-space form.

$$f(x_0) \stackrel{!}{=} 0 \quad (13)$$

Linear approximation of the model (10):

$$\begin{aligned} \dot{T}_{1_{approx}} = f_1(T_{1_0}, T_{\infty_0}, Q_{1_0}) &+ \left. \frac{\partial f_1}{\partial T_1} \right|_{T_{1_0}, T_{\infty_0}, Q_{1_0}} \cdot (T_1 - T_{1_0}) \\ &+ \left. \frac{\partial f_1}{\partial T_{\infty}} \right|_{T_{1_0}, T_{\infty_0}, Q_{1_0}} \cdot (T_{\infty} - T_{\infty_0}) \\ &+ \left. \frac{\partial f_1}{\partial Q_1} \right|_{T_{1_0}, T_{\infty_0}, Q_{1_0}} \cdot (Q_1 - Q_{1_0}) \end{aligned} \quad (14a)$$

$$\dot{T}_{1_{approx}} = f_1(T_{1_0}, T_{\infty_0}, Q_{1_0}) + C_1 \cdot T_{1_{\delta}} + C_2 \cdot T_{\infty_{\delta}} + C_3 \cdot Q_{1_{\delta}} \quad (14b)$$

with

$$T_{1_{\delta}} = (T_1 - T_{1_0}) \quad (15a)$$

$$T_{\infty_{\delta}} = (T_{\infty} - T_{\infty_0}) \quad (15b)$$

$$Q_{1_{\delta}} = (Q_1 - Q_{1_0}) \quad (15c)$$

$$f_1(T_{1_0}, T_{\infty_0}, Q_{1_0}) = \dot{T}_1(T_{1_0}, T_{\infty_0}, Q_{1_0}) = 0 \quad (15d)$$

we define

$$C_1 = \left. \frac{\partial f_1}{\partial T_1} \right|_{T_{1_0}, T_{\infty_0}, Q_{1_0}} = -\frac{UA}{m c_p} - 4 \cdot \frac{\varepsilon \sigma A}{m c_p} \cdot T_{1_0}^3 \approx -9.181 \times 10^{-3} \text{ s}^{-1} \quad (16a)$$

$$C_2 = \left. \frac{\partial f_1}{\partial T_{\infty}} \right|_{T_{1_0}, T_{\infty_0}, Q_{1_0}} = \frac{UA}{m c_p} + 4 \cdot \frac{\varepsilon \sigma A}{m c_p} \cdot T_{\infty_0}^3 \approx 9.181 \times 10^{-3} \text{ s}^{-1} \quad (16b)$$

$$C_3 = \left. \frac{\partial f_1}{\partial Q_1} \right|_{T_{1_0}, T_{\infty_0}, Q_{1_0}} = \frac{\alpha_1}{m c_p} = 5 \times 10^{-3} \text{ K s}^{-1} \%^{-1} \quad (16c)$$

With (16), the single-heater state-space model is:

$$\dot{x}(t) = A_c \cdot x(t) + B_c \cdot u(t) \quad (17a)$$

$$y(t) = x(t) \quad (17b)$$

with

$$A_c = [C_1], \quad B_c = [C_2 \quad C_3], \quad \dot{x} = [T_{1_{\delta}}], \quad u = \begin{bmatrix} T_{\infty_{\delta}} \\ Q_{1_{\delta}} \end{bmatrix}$$

The continuous-time model is then discretised. For a piecewise constant input, the solution of (17) is:

$$x_{k+1} = e^{A \cdot T_{\text{period}}} \cdot x_k + \int_0^{T_{\text{period}}} e^{A \cdot \tau} d\tau \cdot B \cdot u_k \quad (18)$$

With the sample time of the Simulink models

$$T_{\text{period}} = 1 \text{ s}$$

the linearised discrete-time state-space model is:

$$x_{k+1} = A_d \cdot x_k + B_d \cdot u_k \quad (19a)$$

$$y_k = x_k \quad (19b)$$

with

$$A_d = e^{A \cdot T_{\text{period}}} \approx 0.9909, \quad B_d = \int_0^{T_{\text{period}}} e^{A \cdot \tau} \cdot B \approx [9.139 \cdot 10^{-3} \quad 4.977 \cdot 10^{-3}]$$

Task 1, Fig. 3. The figure shows the SKF, EKF and UKF implemented in Simulink for the single-heater model. The model block and all three filters receive the same input vector u : the heater power Q_1 from a pulse generator and a constant ambient temperature (see the operating point (12) and the model (9)). Inside the model block, Gaussian white noise is added to the state to obtain the simulated sensor value T_1 , and every filter receives this same sensor value. The filter outputs should be closer to the model state than the noisy sensor value. Scope 1 in Fig. 3 shows five signals:

- the model state T_1 ,
- the noisy sensor value T_1 ,
- the estimate T_1 of the SKF,
- the estimate T_1 of the EKF,
- the estimate T_1 of the UKF.

The plot of Scope 1 is shown in Fig. 7. Scope 2 compares the posterior covariances of the estimators and is not discussed further.

4.2. Task 2

The dual-heater model is also taken from APMonitor [10]:

$$m c_p \cdot \frac{\partial T_1}{\partial t} = U A_n \cdot (T_\infty - T_1) + \varepsilon \sigma A_n \cdot (T_\infty^4 - T_1^4) + Q_{C12} + Q_{R12} + \alpha_1 Q_1 \quad (20a)$$

$$m c_p \cdot \frac{\partial T_2}{\partial t} = U A_n \cdot (T_\infty - T_2) + \varepsilon \sigma A_n \cdot (T_\infty^4 - T_2^4) - Q_{C12} - Q_{R12} + \alpha_2 Q_2 \quad (20b)$$

$$Q_{C12} = U A_s \cdot (T_2 - T_1) \quad (20c)$$

$$Q_{R12} = \varepsilon \sigma A_n \cdot (T_2^4 - T_1^4) \quad (20d)$$

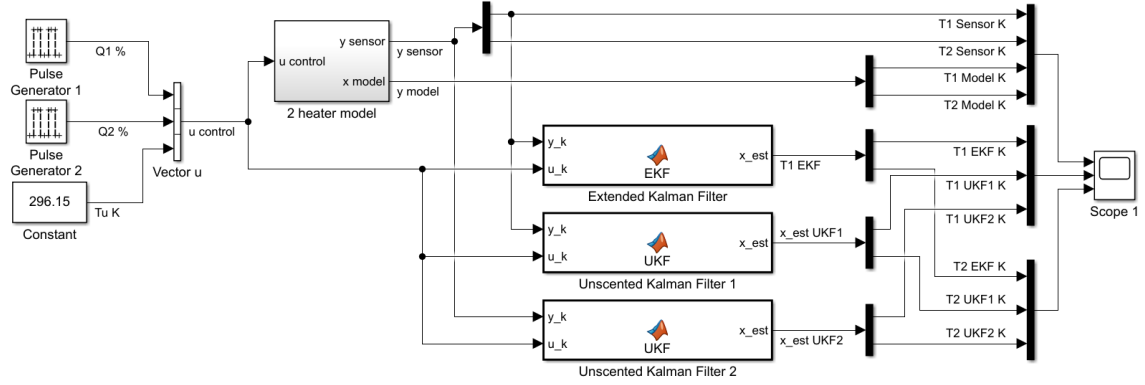


Fig. 4: Task 2, Simulink model *Model_2heater* (screenshot of MathWorks Simulink; not covered by the CC BY license of this report)

Table 1: Model parameters [10]

Symbol	Value	Unit	Symbol	Value	Unit
Q_1	0 to 100	%	Q_2	0 to 100	%
α_1	$10 \cdot 10^{-3}$	$\text{W } \%^{-1}$	α_2	$7.5 \cdot 10^{-3}$	$\text{W } \%^{-1}$
A_s	$0.2 \cdot 10^{-3}$	m^2	A_n	$1 \cdot 10^{-3}$	m^2
m	$4 \cdot 10^{-3}$	kg	c_p	500	$\text{J kg}^{-1} \text{K}^{-1}$
ε	0.9	—	σ	$5.67 \cdot 10^{-8}$	$\text{W m}^{-2} \text{K}^{-4}$
U	10	$\text{W m}^{-2} \text{K}^{-1}$	A	$1.2 \cdot 10^{-3}$	m^2

with the parameters of Table 1.

Task 2, Fig. 4. Task 2 has the same structure as task 1. The single-heater model is replaced by the dual-heater model, and the SKF is replaced by a second UKF. The SKF is not designed for strongly nonlinear models, which is why it is only used in task 1. EKF and two UKFs are therefore compared on the dual-heater model.

All filters and the model receive the same input vector u : the heater powers Q_1 and Q_2 from two pulse generators and a constant ambient temperature (see the operating point (12) and the model (20)). As in task 1, Gaussian white noise is added to the states inside the model block to obtain the sensor values T_1 and T_2 .

EKF and UKF1 estimate T_1 and T_2 from the noisy measurement of T_1 only. UKF2 uses both noisy sensor values T_1 and T_2 to improve the estimates of both temperatures. Scope 1 in Fig. 4 shows ten signals:

- the noisy sensor values T_1 and T_2 ,
- the model states T_1 and T_2 ,
- the estimates T_1 of EKF, UKF1 and UKF2,
- the estimates T_2 of EKF, UKF1 and UKF2.

4.3. Task 3

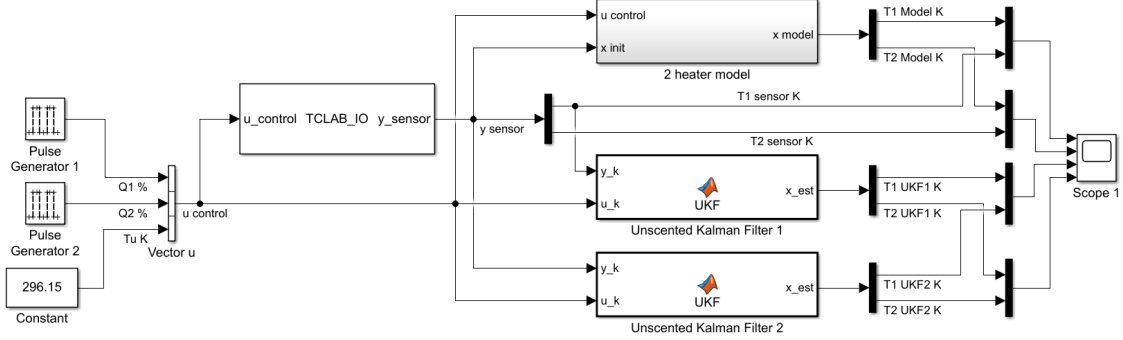


Fig. 5: Task 3, Simulink model *TCLab_2heater* (screenshot of MathWorks Simulink; not covered by the CC BY license of this report)

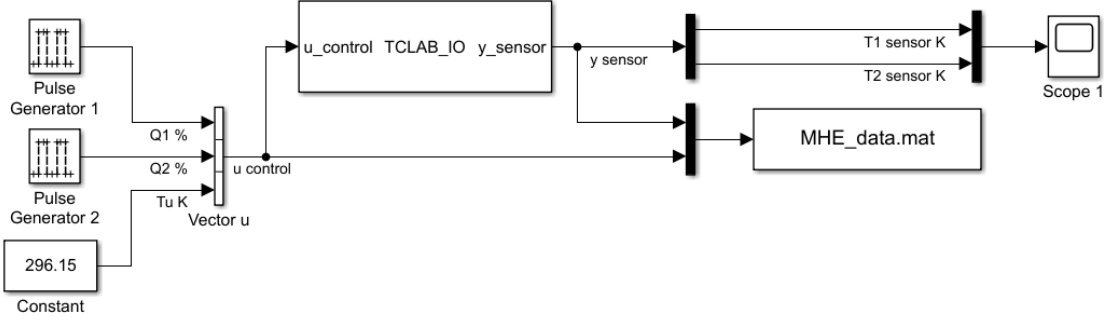


Fig. 6: Task 3, Simulink model *TCLab_MHE_2heater* (screenshot of MathWorks Simulink; not covered by the CC BY license of this report)

Task 3, Figs. 5 and 6. Task 3 consists of the two Simulink models shown above and the MATLAB script *mhe_2heater.m*, which runs the Moving Horizon Estimation. The main difference to task 2 is that the model block is replaced by the real TCLab. When the Simulink model is run for the first time, the COM port of the TCLab has to be specified. The UKFs are implemented in the same way as in task 2.

The MHE is implemented differently from the Kalman filters (Fig. 6). The Simulink model *TCLab_MHE_2heater* writes the input vector u and the measurements T_1 and T_2 to the file *MHE_data.mat*. The MHE is then run offline with the script *mhe_2heater.m*. It uses the measurements T_1 and T_2 and estimates the heat capacity c_p of the heaters.

UKF1 uses the sensor value T_1 and estimates T_1 and T_2 . UKF2 uses both sensor values T_1 and T_2 to obtain better estimates than the raw measurements. The scope in Fig. 5 shows eight signals:

- the noisy sensor values T_1 and T_2 ,

- the estimates T_1 and T_2 of UKF1,
- the estimates T_1 and T_2 of UKF2,
- the model states T_1 and T_2 .

5. Results and Discussion

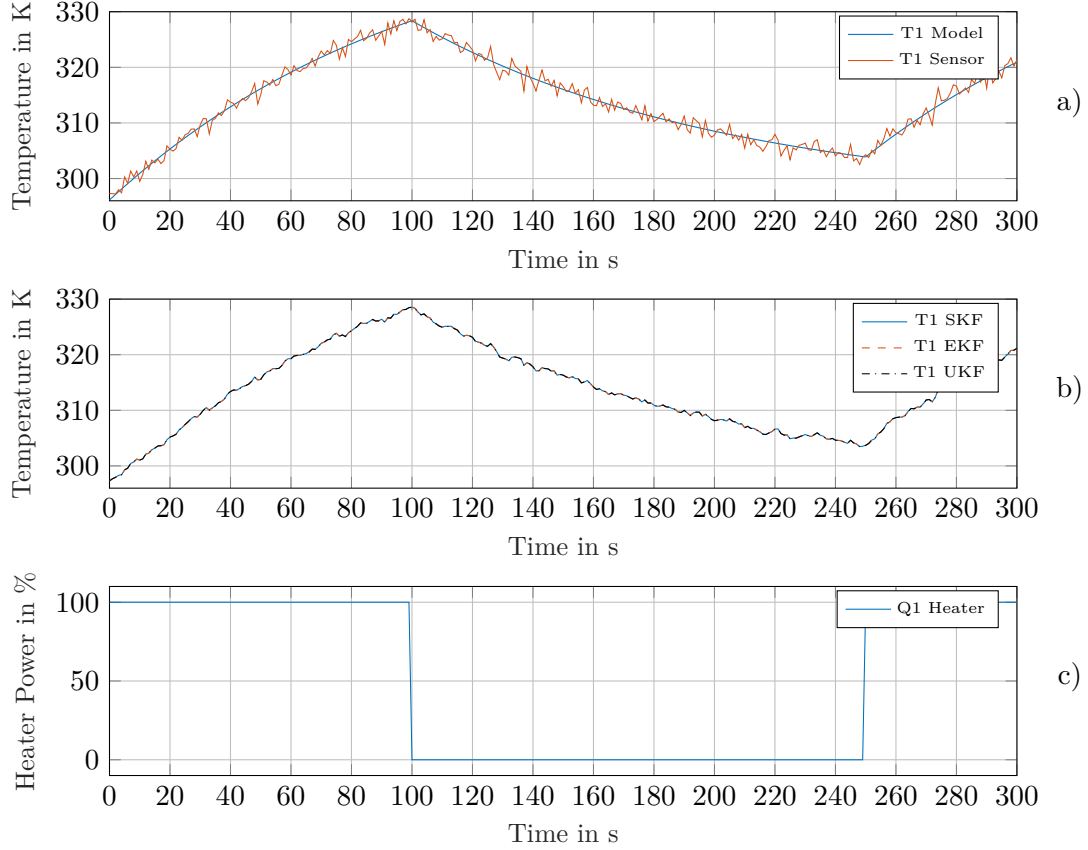


Fig. 7: Task 1, Scope 1 of the Simulink model *Model_1heater*: a) model state and measurement T_1 , b) filter estimates, c) heater power Q_1

Task 1, Fig. 7. Graph a) shows the model state and the measurement of T_1 , graph b) the outputs of the Kalman filters and graph c) the heater power Q_1 . As expected, all filter outputs in graph b) have a lower noise level than the measurement. There is no significant difference between the filters, although the SKF works with the linearised model (10), while EKF and UKF use the nonlinear model (9). A likely reason is the low nonlinearity of the single-heater system around the operating point. In this case, the SKF works as well as the other filters. This does not hold in general for other operating points

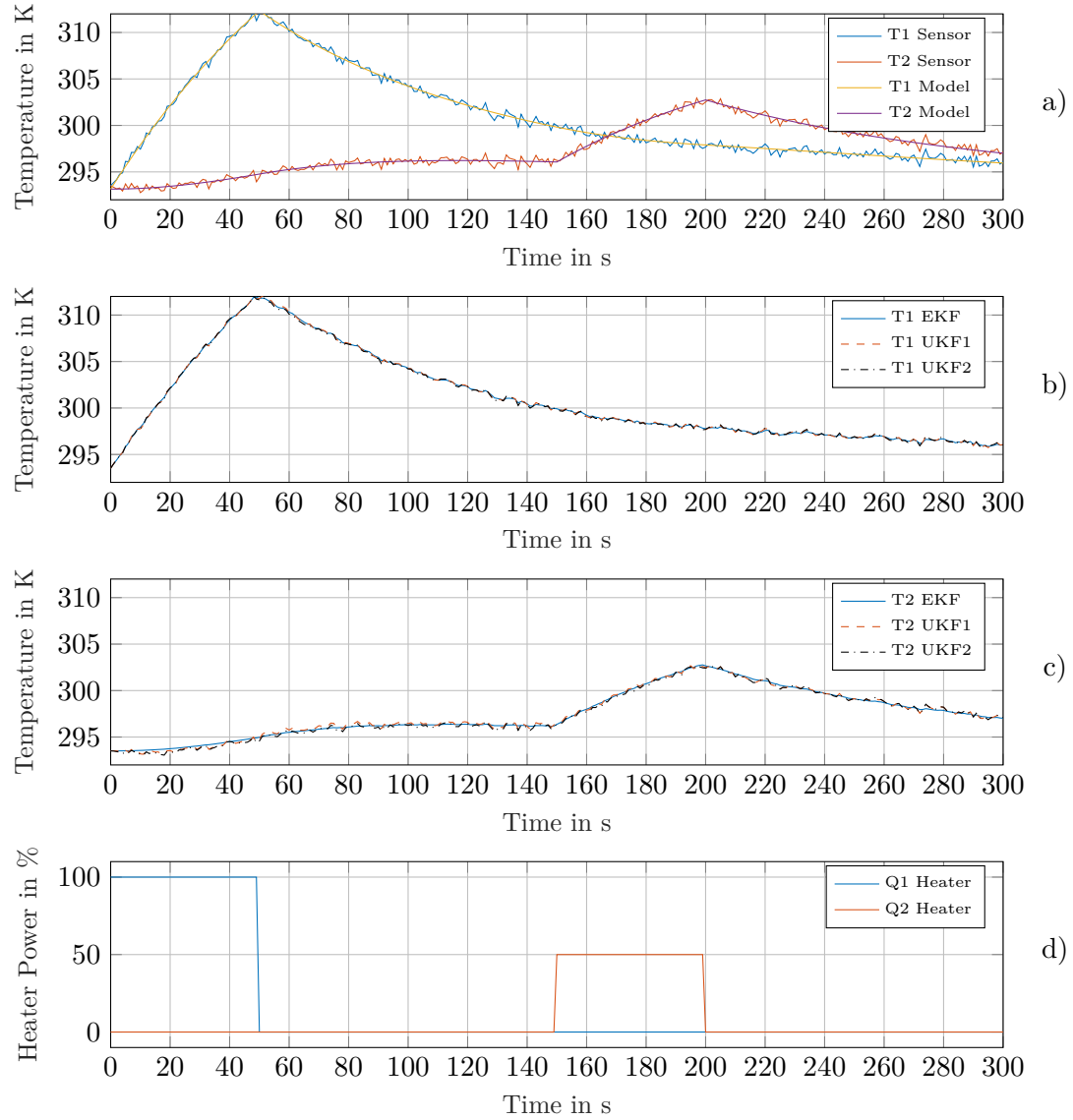


Fig. 8: Task 2, Scope 1 of the Simulink model *Model_2heater*: a) model states and measurements, b) and c) filter estimates of T_1 and T_2 , d) heater powers Q_1 , Q_2

or other systems; whether an approximation is sufficient depends on the requirements of the application.

Task 2, Fig. 8. Graph a) shows the model states T_1 , T_2 and the noisy measurements, graphs b) and c) the outputs of the Kalman filters and graph d) the heater powers Q_1 , Q_2 . UKF1 and EKF estimate T_2 from the measurement of T_1 only. The estimate of T_2 by UKF1 follows the model state T_2 without visible deviation. The filter outputs in graphs b) and c) have a lower noise level than the measurements in graph a). The estimates of the filters are practically equal, although UKF2 uses both measurements T_1 , T_2 and UKF1 only T_1 . A likely reason is that the simulated TCLab and the filters use the same model, so there is almost no model mismatch. This is no longer the case on the real system, as UKF2 in Fig. 9 b) shows.

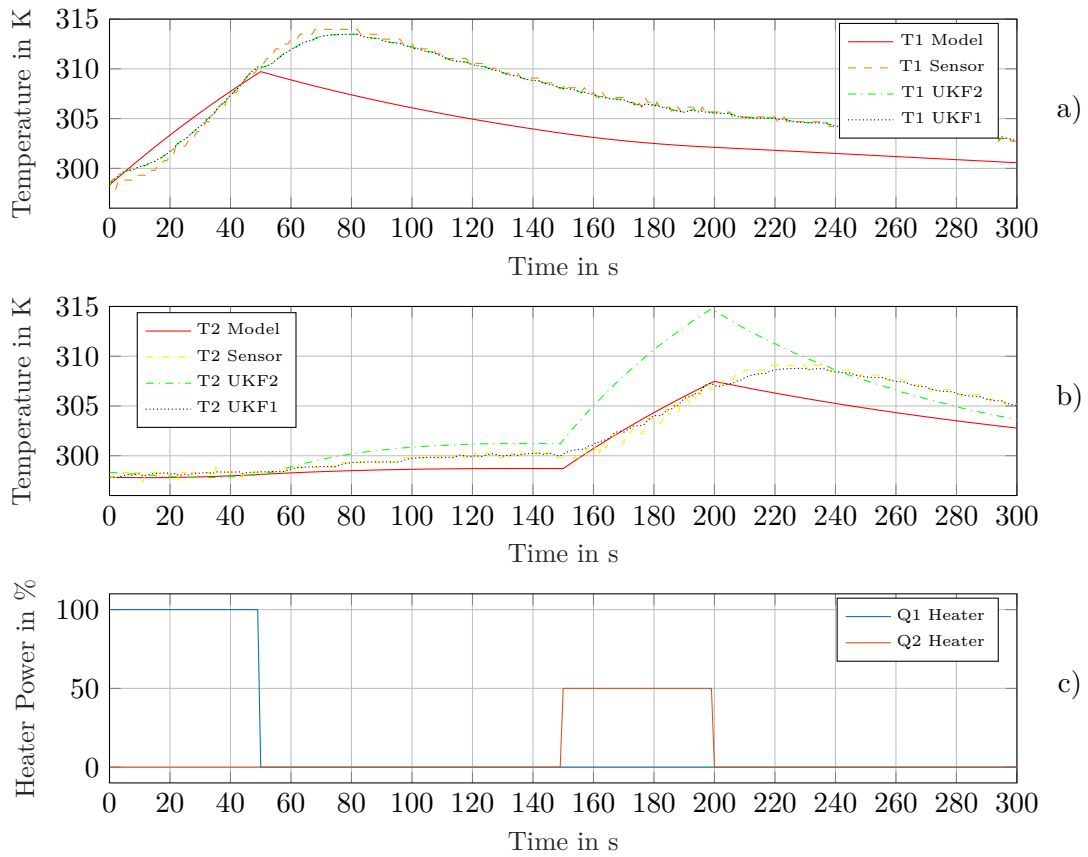


Fig. 9: Task 3, Scope 1 of the Simulink model *TCLab_2heater* (real TCLab): a) T_1 , b) T_2 as model state, measurement and filter estimates

Task 3, Fig. 9. In contrast to Fig. 8, these graphs are recorded on the real TCLab (Fig. 2). Graphs a) and b) compare the model states, the measurements and the filter estimates. UKF1 uses the measurement T_1 and estimates both temperatures T_1 , T_2 ; UKF2 uses both measurements. In graph a), the measurement and both filter estimates follow

roughly the same trajectory. The model state T_1 , however, deviates significantly after the heater power Q_1 is switched off: the measured temperature T_1 keeps rising for a while after the heater has been switched off, and model and measurement only converge again after a long time. In graph b), model and measurement of T_2 behave in the same way.

Several conclusions can be drawn from this. First, the model does not include the heat transfer between transistor, temperature sensor and heat sink, and the heat capacities of the two components are neglected. Second, the temperature sensor and the transistor together have six electrical connections (Fig. 2), which allow heat transfer to and from the circuit board. Third, the heater factor α in the model (9) may not be exact. In summary, the behaviour is caused by model mismatch in several ways.

Within this report, it cannot be proven why the estimate of T_2 by UKF2 in graph b) deviates this far from the measurement of T_2 . It is probably a result of the model mismatch discussed above: UKF1 is implemented in the same way in task 2 and task 3, but graph b) of Fig. 8 does not show this behaviour.

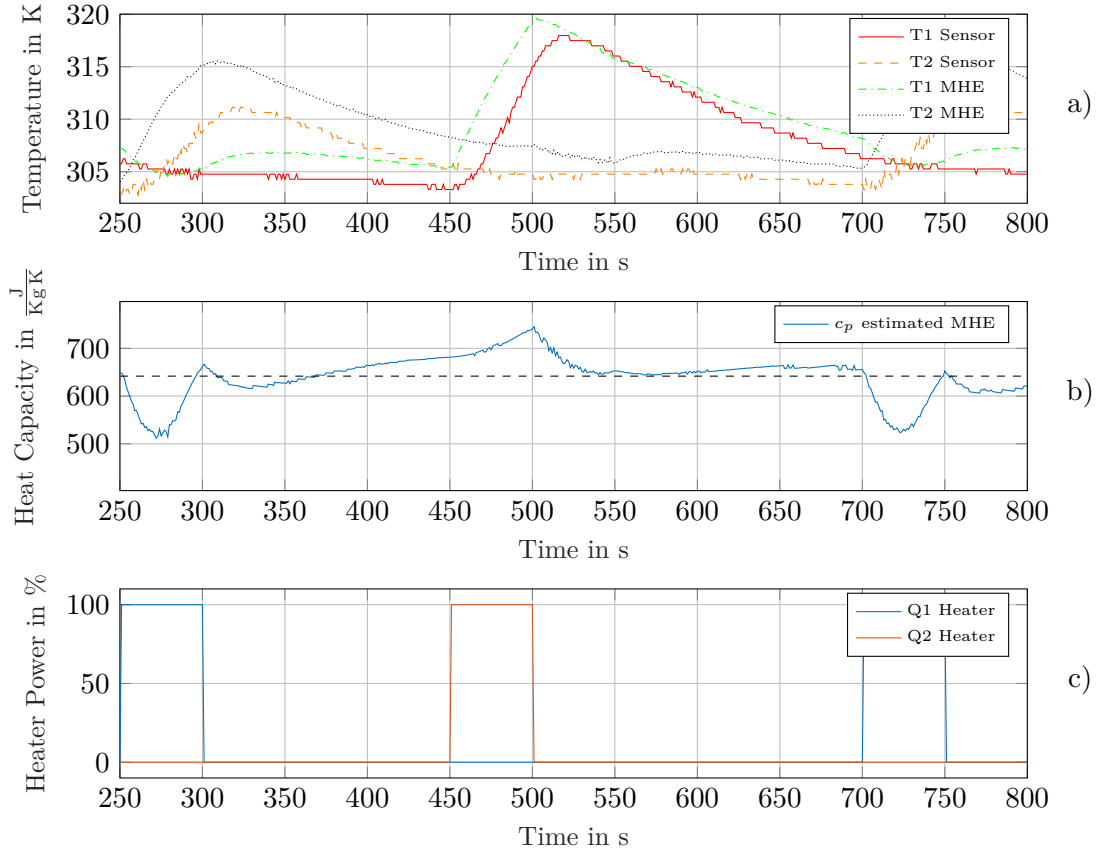


Fig. 10: Task 3, output of *mhe_2heater.m*: a) measured and estimated temperatures, b) estimated heat capacity c_p , c) heater powers

MHE, Fig. 10. Graph a) shows the measured and estimated heater temperatures, graph

b) the estimated heat capacity c_p and graph c) the input. Because of the horizon length, the time axis starts at 250 s. As discussed above, there is a significant model mismatch, and estimating the heat capacity c_p alone may not be the appropriate correction. As graph b) shows, the estimated heat capacity oscillates around approximately $650 \text{ J kg}^{-1} \text{ K}^{-1}$, compared with the nominal value of $500 \text{ J kg}^{-1} \text{ K}^{-1}$.

6. Conclusion

This report applies the Kalman filter and Moving Horizon Estimation to state and parameter estimation on the Temperature Control Lab. In task 1, the standard Kalman filter with a linearised model performs as well as EKF and UKF for the weakly nonlinear single-heater system (Section 4.1). In task 2, the estimation of an unmeasured temperature gives good results as long as the model is exact (Section 4.2). On the real system, a significant model mismatch limits the state estimation (Section 4.3). With an inaccurate model, MHE can estimate parameters and states and thus provide values for correcting the model.

Several topics follow from this work. The SKF should be examined more closely to find where its limit for nonlinear systems lies, for example with a linearised dual-heater model compared on the simulated and the real TCLab. This may also explain why SKF, EKF and UKF show equal trajectories in Figs. 7 and 8. The deviation of the UKF2 estimate of T_2 in Fig. 9 b) may be reduced with a corrected model, e.g. using the parameters estimated by the MHE (Fig. 10). A correction of further parameters such as α_1 and α_2 in the model (20) should also be considered.

About this edition

This is a revised edition (2026) of the course project report submitted in September 2021. The language was edited. In Section 4, the numerical values of the linearised and discretised single-heater model were corrected to the sample time of 1 s used in the Simulink models and to the parameter values of Table 1. The figures of the MHE principle and of the TCLab were replaced by own drawings. The results and conclusions are unchanged.

How to cite

J. Stockhammer, *State Estimation with Kalman Filter and Moving Horizon Estimation on the Temperature Control Lab*, course project report, Otto-von-Guericke University Magdeburg, revised edition, Zenodo, 2026. doi:10.5281/zenodo.22984395

License, trademarks and disclaimer

This report is licensed under the Creative Commons Attribution 4.0 International License (CC BY 4.0), <https://creativecommons.org/licenses/by/4.0/>. Excluded from this

license are all third-party contents: the screenshots of Simulink models in Figs. 3, 4, 5 and 6 contain user interface elements of software by The MathWorks, Inc. Any reuse of such contents beyond the statutory right to quote requires the permission of the respective rights holder.

MATLAB and Simulink are registered trademarks of The MathWorks, Inc. Arduino is a trademark of Arduino SA. The Temperature Control Lab (TCLab) is a product of APMonitor. All other product names mentioned are trademarks or registered trademarks of their respective owners and are used only to identify the products. This report is not affiliated with or endorsed by these companies.

This report is a course project report and has not been peer reviewed. It is provided as is, without warranty of any kind. The author accepts no liability for any use of the methods, code or data described.

References

- [1] R. E. Kalman, A new approach to linear filtering and prediction problems, *Journal of Basic Engineering* 82 (1) (1960) 35–45. doi:10.1115/1.3662552.
- [2] R. Burkey, Virtual AGC: Luminary, accessed 2021 (2021).
URL <http://www.ibiblio.org/apollo/Luminary.html>
- [3] C. Baumann, Das Kalman filter wird 50 (2008).
URL https://www.convict.lu/htm/rob/Das%20Kalman%20Filter%20wird%2050_v_1.4.pdf
- [4] R. Dürr, Kalman filter for linear systems, lecture notes, course State Estimation, Otto-von-Guericke University Magdeburg (2020).
- [5] R. Dürr, Extended Kalman filter for nonlinear systems, lecture notes, course State Estimation, Otto-von-Guericke University Magdeburg (2020).
- [6] R. Dürr, Unscented Kalman filter for nonlinear systems, lecture notes, course State Estimation, Otto-von-Guericke University Magdeburg (2020).
- [7] R. Marchthaler, S. Dingler, *Kalman-Filter: Einführung in die Zustandsschätzung und ihre Anwendung für eingebettete Systeme*, Springer Vieweg, Wiesbaden, 2017. doi:10.1007/978-3-658-16728-8.
- [8] E. L. Haseltine, J. B. Rawlings, Critical evaluation of extended Kalman filtering and moving-horizon estimation, *Industrial & Engineering Chemistry Research* 44 (8) (2005) 2451–2460. doi:10.1021/ie034308l.
- [9] do-mpc developers, Basics of moving horizon estimation, do-mpc 4.0.0 documentation, accessed 2021 (2021).
URL https://www.do-mpc.com/en/latest/theory_mhe.html

- [10] J. Hedengren, Temperature control lab, APMonitor, accessed 2021 (2021).
URL <https://apmonitor.com/pdc/index.php/Main/ArduinoTemperatureControl>
- [11] J. Hedengren, Arduino setup and installation, APMonitor, accessed 2021 (2021).
URL <https://apmonitor.com/pdc/index.php/Main/ArduinoSetup>
- [12] J. Stockhammer, State estimation with Kalman filters and MHE: script and recorded data (2026).
URL <https://github.com/josto-me/kalman-filter-state-estimation>